

Characterising the Plasma Focus Pinch and Speed Enhancing the Neutron Yield

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ABSTRACT

From dimensional considerations and also experiments the compressed radius, pinch length and ‘lifetime’ of the plasma focus pinch are each proportional to the anode radius ‘a’. Further an analysis of a range of plasma focus machines shows that the speed parameter $(I/a)/\rho^{1/2}$, where I is the driving current and ρ is the ambient density, has practically the same value (90kA m⁻¹ per torr^{1/2} in deuterium, variation of less than 10%) over the whole range of small to big machines. This indicates that the speed (v) and energy density (hence plasma temperature T) have practically the same values over the range of machines. This gives us the well-known $Y \sim I^4$ neutron yield law that is observed. This observed law is due to the fixed speed operation over the whole range of these machines. If the value of (I/a) is allowed to increase, speed will increase with (I/a) and the yield scaling will change $Y_{th} \sim I^4(I/a)^4$ or $I^4 v^4$ giving rise to the concept of speed enhancement of neutron yield.

Plasma focus scaling, Neutron yield enhancement, Plasma Focus computation

I: The Speed Parameter $S: (I_p/a)\rho^{-1/2}$

Using a snowplow model for the axial phase and a slug model for the radial phase (Fig.1), simply by normalizing equations, characteristic axial & radial transit times for the plasma shock are respectively [1-4]:

$$t_a = \left[\frac{4\pi^2(c-1)}{\mu \ln c} \right]^{1/2} \frac{z_0 \rho^{1/2}}{(I/a)} \quad (1)$$

$$t_r = \left[\frac{16\pi^2}{\mu(\gamma+1)} \right]^{1/2} \frac{a \rho^{1/2}}{(I/a)} \quad (2)$$

where $c = b/a$, b is the outer radius of the coaxial electrodes, a is inner radius, μ is permeability, z_0 is length of inner electrode, I is characteristic current, ρ is ambient density and γ is the specific heat ratio.

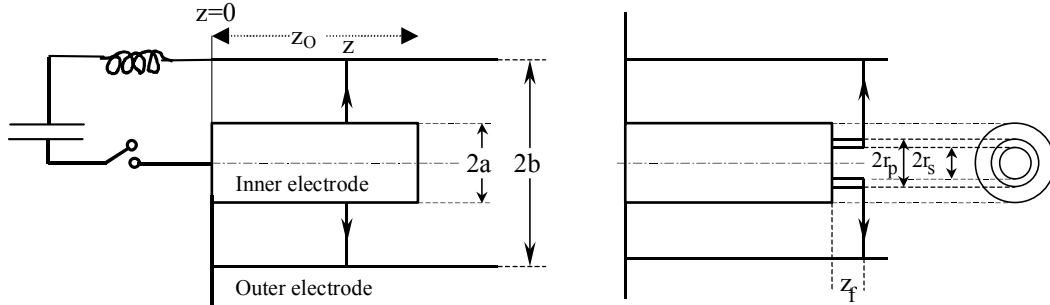


Fig. 1. Schematic of the dynamics of the plasma focus for the 1-D computational model: (a) axial phase - snowplow model; (b) radial phase - slug model

From these we write [5] the characteristic speeds for the axial and radial shock, respectively as:

$$c_a = \frac{z_o}{t_a} = \frac{(I/a)}{\rho^{1/2}} \left[\frac{\mu \ln c}{4\pi^2 (c-1)} \right]^{1/2} = \frac{(I/a)}{\rho^{1/2}} g_a \quad (3)$$

$$c_r = \frac{a}{t_r} = \frac{(I/a)}{\rho^{1/2}} \left[\frac{\mu(\gamma+1)}{16\pi^2} \right]^{1/2} = \frac{(I/a)}{\rho^{1/2}} g_r \quad (4)$$

where g_a and g_r are the effective geometrical constants governing the axial and radial phases, respectively; g_r is a constant and g_a varies by less than 10% from a mean value of 0.52 over the range of $c = 1.5$ to 2. Hence c_a and c_r and the operational speeds in the axial and radial phases depend only on the speed parameter S: $(I_p/a)\rho^{-1/2}$ where I_p is the peak current entering the focus tube.

II: Scaling: Focus pinch length, radius and lifetime, each dependent on anode radius ‘a’

A radial collapse model which has proven useful for comparison with experimental work is as shown in Fig. 1b. Essentially a zero length pinch starts at $r = a$ and elongates as it pinches inwards. The collapsing magnetic piston at position r_p pushes ahead of it a radially converging structure at position r_s . Using this model and having shown that the characteristic speed of radial transit is the same over the range of devices, it follows that the compression time $t = t_{\text{comp}}$ (see Fig. 2) depends on the radius a of the anode. As the shock converges on the axis we may define the position of the piston at this time as r_c . After the shock hits the axis a reflected shock propagates outward. In the meantime the piston (see Fig. 2) continues inwards until it is hit by the reflected shock at which point the radially inward motion of the piston is reversed at position $r_p = r_{\text{min}}$. Experimentally it is observed that the plasma column usually pinches in a second time sometimes to an even smaller radius before finally exploding outward. This ends the pinch phase and defines the quantity t_{pf} as shown in Fig. 2..

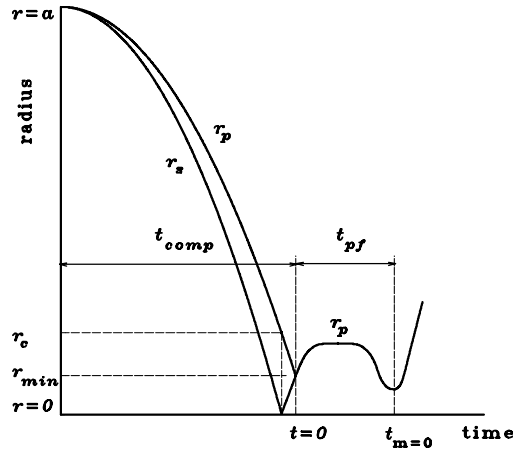


Fig 2: Schematic of the radial collapse

A numerical computation of this model [1-4, 7] shows that both r_c and r_{min} are proportional to anode radius ‘a’. The ending of the pinch phase is observed to be caused by instabilities. Thus we may relate the period t_{pf} to the transit time of small disturbances across the compressed plasma column. The speed of these small disturbances is practically a constant over the range of devices because of the near constancy of plasma temperature, density and magnetic flux. Thus in such a scenario t_{pf} also depends on the radius a . We have carried out a numerical computation with end-point fixed by an energy-balance consistent method [8,9] and found the following for the deuterium focus and the neon focus:

		Deuterium	Neon (for SXR)
minimum radius	r_{min}	0.12a	0.04a
maximum length	Z	0.8a	0.8a
radial shock transit	t_{comp}	$5 \times 10^{-6}a$	$4 \times 10^{-6}a$
pinch lifetime	t_f	$2 \times 10^{-6}a$	$1 \times 10^{-6}a$

where, for the times in sec, the value of anode radius, a, is in m.

Similar results are applicable to the gross pinch of the plasma focus in other gases such as Argon, Neon and Xenon. Generally for these other gases $r_{\min} < 0.1a$. The difference in the computation involves the value of γ . Scaling these parameters of the focus to the anode radius a is extremely useful for understanding and for applications. We have carried out exhaustive series of experiments [7,10], using TEA nitrogen laser nanosecond shadowgraphy (see Fig 3), which confirm these theoretical results. Agreement is also observed with more recent detailed computation [1].

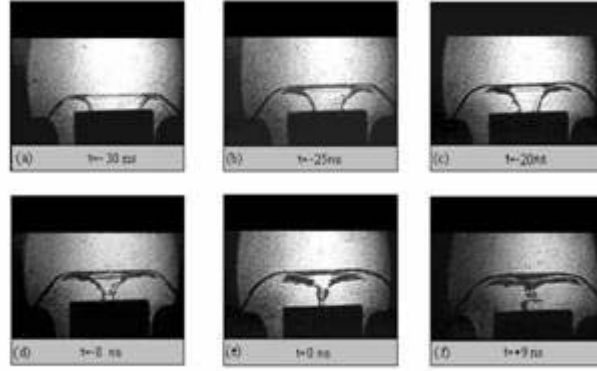


Fig 3. Sequence of compression in UNU/ICTP PFF at 4.0 mbar deuterium [20]

III: Drive parameter S is constant over range of machines

We next return to discuss the drive parameter S: $(I_p/a)p^{-1/2}$. We had surveyed [5] the drive parameter for a range of machines, from large to small, as compiled in Table 1.

Device/Author [Reference]	V (kV)	W (kJ)	p (torr D ₂)	I_p (kA)	a (cm)	$I_p/a/p^{1/2}$ (kA/cm/[torr D ₂] ^{1/2})
DPF-78 [24]	60	28	5	550	2.5	98.4
SPEED 2 [25]	183	70	4	1000	6	83.3
PF-360 [18]	21	60	3	750	5	86.6
Mather [19]	17.3	13.5	6.5	537	2.5	84.3
UMPF [20]	14	6	6.5	300	1.25	94.1
UNU/ICTP-PFF [23]	14	2.9	3.5	172	0.95	96.8
Bernstein [21]		27	8	820	3.9	74.3
POSEIDON [22]	60	280	3.8	1200	6.55	94.0

Table 1 Actual-shot data of some neutron-optimized Mather-type plasma focus devices and their calculated drive parameter. The mean value and the corresponding standard deviation of the drive parameter are 89.0 and 7.7 kA/cm/(torr D₂)^{1/2} respectively.

In this compilation care is taken that the data represent actual neutron-optimized shots as published. Actual-shot data needs to be distinguished from machine data which may give maximum current capability of a machine or design current which when combined with the pressure of an actual shot may give a misleading drive parameter since the actual shot is usually carried out not at the maximum or designed current. Thus much published data of machines could not be used in this compilation because of uncertainty over actual shot data. In Table 1 the value of the peak current I_p flowing into the focus tube is used as this value is more usually quoted than the plasma current and thus using the peak current more actual shot data points are available.

Table 1 shows that the drive parameter S for a range of neutron-optimized plasma focus devices is 89±8 kA/cm per (torr)^{1/2} of deuterium. The standard deviation of less than 10% is quite remarkable since the machines in the

compilation range from 3 to 280 kJ and include early machines of Mather and Bernstein, advanced machines such as SPEED 2 and small training machines.

Any modeling based on electromagnetic drive such as we have shown in Section I will show that the drive parameter S determines the speed in both the axial and radial phases [2,3]. Having the same value of drive parameter S indicates that the range of machines surveyed all operate with the same axial and radial speeds. This is not contradictory to the oft-reported speeds of almost 10 cm/μs for the axial phase just before the radial phase and a radial speed of about 25 cm/μs when the imploding shock is nearing the axis. Since the energetics of the plasma focus is determined by the shock speed up to the time the radial shock goes on axis, constancy of the drive parameter gives a strong indication that the temperature is also the same over the range of plasma focus surveyed. The density also varies remarkably little, a range of 2, considering the large range of energies encompassed in the survey. Thus the plasma energy density of the machines has also relatively small variation. The mechanisms for radiation of energy and particles depend ultimately on the energy density and/or the temperature. The small variations observed in these quantities over the large range of machines whilst on the one hand is an advantage for comparing experiments may perhaps on the other hand act as a severe limitation.

IV: SCALING OF YIELD

For a thermalised plasma we can generally assign density n and temperature T. If the particles of density n interact among themselves with a cross section C(T), generally dependent on T, we may write down a general radiation yield, Y, relationship as follows:

$$Y \sim n^2 (\text{volume}) (\text{lifetime}) C \quad (5)$$

For the plasma focus, given the dimensional and temporal dependence as discussed in Section II we have:

$$Y \sim n^2 a^4 C \quad (6)$$

Constant S operation

Given the observed constancy in S we have: $I \sim ap^{1/2}$ (7)

Hence applying equation (7) to Eq (6) with $n \sim \rho$ and noting that since $T = \text{constant}$ over the range of devices, the cross section C is also constant and we have /9/:

$$Y \sim I^4 \quad (8)$$

This yield law really applies to the thermonuclear neutron yield. Experimentally for a small neutron-optimised plasma focus it has been shown [11] that the thermonuclear component of the neutron yield is only 15%, the other 85% being ascribed to beam-target mechanisms. Nevertheless we note that these yield proportions are occurring at a plasma ion temperature of about 1 keV and a beam energy of about 50 keV.

V: SPEED-ENHANCEMENT OF NEUTRON YIELD

We now discuss the concept of speed enhancement [12,6]. If the driver parameter S is increased, the drive energy density increases, the speed increases and we may expect the focus pinch ion temperature also to increase. This leads to a dramatic increase in neutron yield since in the range of 1 – 10 keV, the neutron fusion cross-section is a rapid function of T with $C \sim T^n$, n being greater than 4. We note $T \sim v^2$ where v is the driven plasma shock speed. If we keep 'a' constant as the drive current is increased, we increase the drive parameter S. This will give us a thermonuclear yield component better than

$$Y \sim I^8 \quad (9)$$

Increase in speed will also increase the energy of the beam component. The beam component will however not have a significant increase since above 50 keV the cross section C barely increases with T. This means that as operational speed is increased, the plasma focus neutron yield becomes more thermonuclear.

Operating the focus at higher velocities, hence higher temperatures, the thermonuclear component Y_{th} of the total neutron production is enhanced. A more general analysis shows that

$$Y_{th} \sim I^4 v_{axial}^4 \quad (10)$$

and the beam-target component of the neutron yield

$$Y_{b-t} \sim I^{4.5} v_{axial}^{-1.5} \quad (11)$$

The total neutron yield is $Y_n = Y_{th} + Y_{b-t}$. If the experiments do not limit the speed of operation, then Y_{th} will increase above the I^4 scaling whilst Y_{b-t} will have a scaling law poorer than $I^{4.5}$. In the limit keeping the anode radius constant whilst increasing the current,

$$Y_{th} \sim I^8 \text{ and } Y_{b-t} \sim I^3$$

VI: Experimental considerations

However it is not a simple matter to increase the drive speed. Experiments in deuterium have shown that if the peak axial speed is pushed above 10 cm/ μ s, plasma focus quality simply deteriorates. A force-field flow-field separation mechanism [13] has been proposed to account for the deterioration of the focus quality. At speeds above 5 cm/ μ s the specific heat ratio γ of the heated deuterium plasma [16] rises rapidly towards 5/3. The shock layer thickness is extremely sensitive to γ and thickens above these speeds. Moreover above these speeds the Magnetic Reynolds Number (MRN) is in a crucial transition above 1. The transition from low MRN to high MRN with increasing speed is extremely rapid as, in this ionizing regime, the MRN is proportional [17] to (speed)⁴. Thus as the speed increases from a crucial 5 cm/ μ s, say, to 10 cm/ μ s, the MRN would have transitioned from a value of 1 (poor electromagnetic coupling) to 16 (good electromagnetic coupling). This latter case provides a separation between the force field and the mass field it is driving; a separation which increases with distance traveled at speeds in the region of 10 cm/ μ s and above. Whilst such a separation is conducive to energy transfer, at the end of the axial phase, as the mass-force system expands from the axial to the radial phase, the force field is left behind and is still pushing axially whilst the radial shock is already imploding inwards. As this separation increases the quality of the radial focus pinch deteriorates. There are indications of such a mass field-force field separation at the higher speeds from schlieren photographs (see Fig 4).

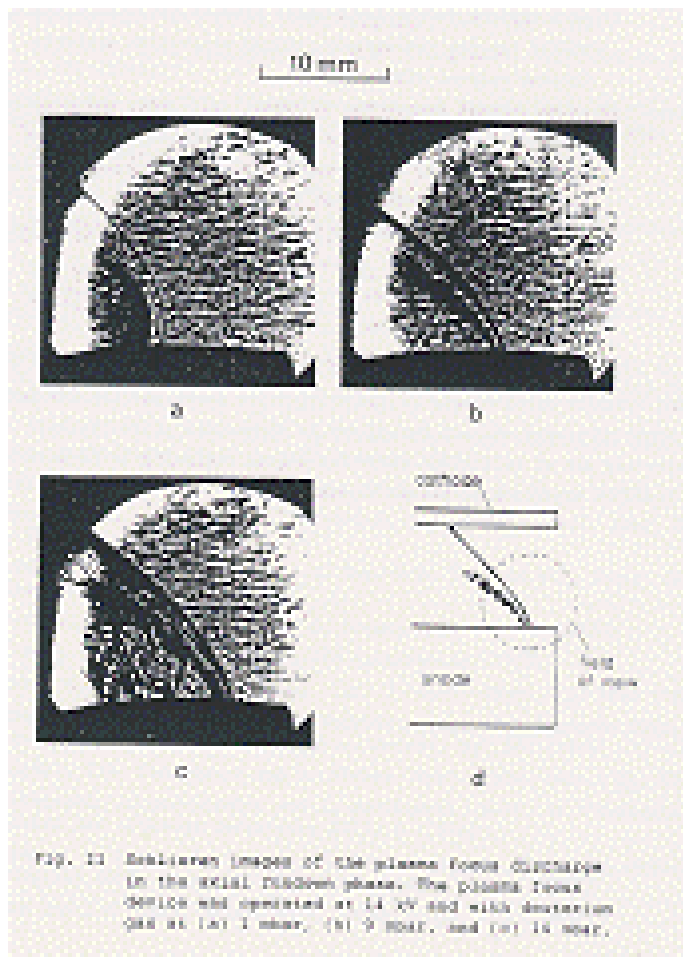


Fig 4 Schlieren pictures showing current sheet separation from shock front, indicating increasing force field-mass field separation at increasing speeds

Stepped Anode for Speed Enhancement whilst minimizing Separation

An experiment [7] was carried out to achieve this speed increase yet overcome the separation effect by keeping the ‘speed-enhanced’ region short. A stepped-anode, Fig 5, was used consisting of a ‘normal’ radius section designed with the normal S value, followed by a short ‘speed-enhanced’ section for which the S value was increased by a reduction of ‘a’. By this means speeds up to 15 cm/ μ s was achieved with reasonably consistent focussing. Neutron-yield enhancement [7, 14] was indicated (see Fig 6). SXR yield in DEUTERIUM was also speed-enhanced [15]

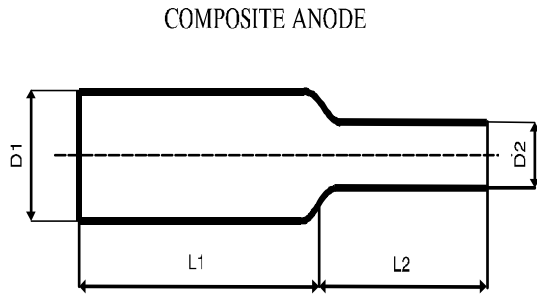


Fig 5. Stepped anode

Argon in Axial Phase Collapsing onto Deuterium in Radial Phase

Another method would be to use argon or krypton in the axial phase collapsing onto a deuterium gas (e.g. jetting out of a hollow anode). Even at speeds of 15 or 20 cm/ μ s the argon or krypton will not be fully ionized, hence the axial shock will have a specific heat ratio γ close to 1, with the piston moving almost together with the shock, thus reducing the mass field-force field separation. In this manner could be obtained much higher focus collapse speed in deuterium, with attendant speed-enhanced neutron yield.

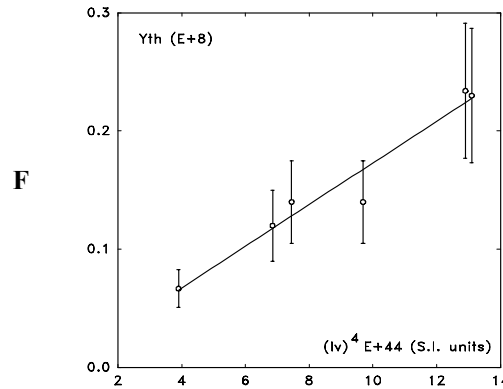


Fig 6. Non-beam component of neutron yield as function of $(Iv)^4$, speed-enhanced UNU/ICTP PFF, operated at 14 kV, 4.5 mb

REFERENCES

1. S Lee "Plasma Focus (Radiative) Computation Model and Code; <http://sci.nie.edu.sg/ckplee>; e-published by CKP Lee, NTU/NIE-first published June 2000
2. S Lee et al "Technology of the Plasma Focus" in *Laser and Plasma Technology*, Ed by S Lee et al, World Sci Pub Co, Singapore (1985) p387-420
3. S Lee "Plasma Focus Model Yielding Trajectory and Structure" in *Radiation in Plasmas* Ed B.Mc Namara, World Sci Pub Co (1984) 978-987.
4. S Lee "Technology of a small plasma focus-incorporating some experience with UNU/ICTP PFF" in *Small Plasma Physics Experiments II*, Ed S Lee & P H Sakanaka, World Sci Pub Co (1989) p113-169
5. S.Lee & A.Serban. Dimensions & Lifetime of Plasma Focus Pinch. IEEE Trans Plasma Sc. USA (1996) 1101-1105.
6. S Lee & A Serban, "Speed-enhanced Neutron Yield in Plasma Focus", International Conf on Plasma Phys, Foz do Iguassu, Brazil, Oct 1994, Conference Procs Vol 1, p181-184
7. A Serban, "Anode Geometry & Focus Characteristics" PhD thesis, NTU (1995)
8. S Lee, "An Energy-consistent Snow-plough Model" J Phys D: Appl Phys 16 (1983) 2463-2469
9. S Lee "Density Ratios in Compressions Driven by Radiation Pressure" Laser & Part Beams, 6, (1988) 597-606
10. M H Liu "Soft X-rays from Compact Plasma Focus", PhD thesis, NTU (1997)
11. S P Moo, C K Chakrabarty & S Lee, IEEE Trans Plasma Sci **19**, 515-519 (1991)
12. S. Lee. Sharing of fusion-related technology among developing countries. Invited pp. Energy Independence Conf 1987; Rio de Janeiro; Fusion Energy & Plasma Physics, World Scientific (Ed: P.H. Sakanaka) (1988). 754-781.
13. S Lee Technology of a Small Plasma Focus in "Small Plasma Experiments II" Ed S Lee & P H Sakanaka, World Scientific Publishing Co (1990)
14. A Serban & S Lee "Experiments on Speed-Enhanced Neutron Yield" J Plasma Phys (1998) 60(1):3-15
15. A Serban & S Lee "Soft x-ray Emission from a Small Plasma Focus" Plasma Sources S & T (1997) 16:25-36
16. S H Saw, S Lee & C S Wong "A Numerical Study of the Effect of γ on Shock Speed for Atomic Hydrogen and Molecular Hydrogen" J Fizik Malaysia 9 (1988) 93-97
17. S Lee, S H Saw, AA Elfetouri, A M Shengher, E O Shumaki & N A Hassan "Magnetic Reynolds Number and Current Sheet Structure" J Fizik Malaysia 12 (1991) 25-29
18. S. Czekaj, A. Kasperczuk, R. Miklaszewski, M. Paduch, T. Pisarczyk, and Z. Wereszczynski, "Diagnostic Method for the Magnetic Field Measurement in the Plasma Focus Device", *Plasma Physics and Controlled Fusion*, **31**, 4, p. 587-594, 1989
19. J.W. Mather & P.J. Bottoms, "Characteristics of Dense Plasma Focus", *Phys. Fluids*, **11**, 3, p. 611-618, 1968

- 20 K.H. Kwek, T.Y. Tou, and S. Lee, "Current Sheath Structures of the Plasma Focus in the Run-Down Phase", *IEEE Trans. on Plasma Sci.*, **18**, 5, p. 826-830, Oct. 1990
- 21 M.J. Bernstein and G.G. Comisar, "Neutron Energy and Flux Distributions from a Crossed-Field Acceleration Model of Plasma Focus and Z-Pinch Discharges", *Phys. Fluids*, **15**, 4, p. 700-707, April 1972
- 22 U. Jager, "Über die Reaktionsmechanismen im Plasmafokus - Untersuchen an beschleunigten Deuteronen und Reaktionsprotonen", Univ. Stuttgart, IPF-86-1, 1986
- 23 S. Lee, T.Y. Tou, S.P. Moo, M.A. Eissa, A.V. Golap, K.H. Kwek, S. Mulyodrono, A.J. Smith, Suryadi, W. Usada, and M. Zakaullah, *Am. J. Phys.* **56**, 1, p. 62-68, 1988
- 24 H. Schmidt, "Generation, Physics and Applications of Superdense Hot-Z-Pinch Plasmas", Rep. on activities, March '92 - Feb. '93, Univ. Stuttgart, Germany (Contr. SCI CT91-0728 (TSTS))
- 25 G. Decker, W. Kies, M. Malzig, C. van Caler, and G. Ziethen, "High Performance 300 kV Driver Speed 2 for MA Pinch Discharges", *Nuclear Instruments and Methods in Physics Research A*249, p. 477-483, 1986